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31 MARCH 1953
SUPERSEDING
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5 February 1952

FEDERAL SPECIFICATION

BATH, WHIRLPOOL, CRM.

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—The whirlpool baths covered by this specification are the arm type or the leg type with electric turbine ejector.

1.2 Classification.

1.2.1 *Type, grade, and sizes.*—The whirlpool baths covered by this specification shall be of one type and grade and of the following sizes, as specified in the invitation for bids:

Size 1.—Arm whirlpool bath
Size 2.—Leg whirlpool bath

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

2.1 Specifications.—The following Federal Specification, of the issue in effect on date of invitation for bids, forms a part of this specification:

QQ–S–766—Steel, Corrosion-Resisting Plates, Sheets, Strips, and Structural Shapes.

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by check, money order, cash, or Government Printing Office coupons, to the General Services Administration, Business Service Center, Region 3, Seventh and D Streets, S. W., Washington 25, D. C. This office will also honor deposit numbers issued by the Government Printing Office. Single copies of product specifications required for bidding purposes are available without charge at the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, Los Angeles, Seattle, and Washington, D. C. Prices of additional copies may be obtained from the GSA Regional Offices.)

2.2 Other publications.—The following publication, of the issues in effect on date of invitation for bids, forms a part of this specification:

National Board of Fire Underwriters Publication:

NBFU Pamphlet No. 70—National Electrical Code, Standard of the National Board of Fire Underwriters for Electrical Wiring and Apparatus as amended.

(NBFU standards may be obtained from the National Board of Fire Underwriters, 85 John Street, New York 7, New York; 222 West Adams Street, Chicago 6, Illinois, and Merchants Exchange Building, San Francisco, California.)

2.3 Specifications and standards applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Component parts.

3.1.1 *Whirlpool arm bath.*—Whirlpool arm bath shall consist of a tank mounted on a pedestal, an electric turbine ejector, turbine elevator, thermostatic water-mixing valve assembly, dial thermometer, switch box, drain, and overflow valve.

3.1.2 *Whirlpool leg bath.*—Whirlpool leg bath shall consist of a floor-mounted tank, electric turbine ejector, turbine elevator,

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thermostatic water-mixing valve assembly, dial thermometer, switch box, drain, and overflow valve.

3.2 Materials.

3.2.1 *Sheet metal.* — Sheet metal shall be corrosion-resisting steel of thickness specified and shall conform to class 1 of Federal Specification QQ–S–766. Commercial tolerances shall apply on thickness of sheet metal.

3.2.2 *Pipes, valves, and fittings.* — Pipes, valves, and fittings shall be of brass or bronze coated with polished chromium plating over nickel.

3.2.3 *Welding.*—Tank welds shall be continuous. Welds shall be thorough, nonporous, free from cracks and other imperfections. All welds shall be cleaned. All exposed welds shall be ground and polished to blend with the adjacent surfaces.

3.3 Construction.

3.3.1 *Tanks.*—Tanks shall be constructed of corrosion-resisting steel not less than 0.050 inch thick. The tank shall be oval in shape and the top edge shall be finished with a beaded rim, neatly rolled over a ⅜-inch-diameter corrosion-resisting steel rod. The tank shall be constructed with continuous welds. After welding, the tank and all its seams shall withstand a test for leakage by applying a minimum of 15 pounds per square inch air pressure to the inside of the tank. The crevice of the joints between the bottom and the side wall on the inside of the tank shall be filled with a small smooth fillet of 50-50 solder. The bottom shall be so constructed as to prevent buckling or sagging from the weight of the water, or by other means, and shall be securely mounted. The method of constructing and attaching the bottom shall be such that no free play can be obtained by the intermittent pressure of water or by other means. The bottom of the tank shall have a minimum pitch ¼ inch sloping downward to the waste opening in order to insure complete drainage of the tank.

3.3.2 *Size 1, arm whirlpool bath.* — The inside measurements of oval tank shall be approximately 18 inches deep, 15 inches wide, and 28 inches long. The side wall of tank shall have a 1-inch apron finished at its lower edge with a tight double hem. A pedestal attachment base with four mounting bolts shall be recessed into, and welded to, the tank apron. The pedestal base shall be approximately 8 inches in diameter and provided with four holes for floor mounting. The base shall accommodate a substantial tubular upright of sufficient height to allow tank mounting with the top rim approximately 36 inches from the floor. Pedestal shall be made of brass or bronze, chromium over nickel plated, or of corrosion-resisting steel, satin finish.

3.3.3 *Size 2, leg whirlpool bath.* — The inside measurements of the oval tank shall be approximately 28 inches deep, 20 inches wide, and 36 inches long. The over-all height of the tank shall be 33 inches, including a 5-inch apron extending from bottom of tank to the floor for supporting the tank. The bottom rim of the apron shall have a tight double hem approximately 1 inch wide on the outside of tank. The bottom of the tank shall be so constructed as to prevent buckling or sagging from weight of water or patient. The bottom of the tank shall be creased or pitched to the drain to provide complete drainage.

3.4 Tank fittings.—One end of the tank shall be equipped with a direct-connected waste and overflow having an outside dimension of 2 inches. The waste line for size 1 shall include a combination drain valve and overflow, trap with union fittings to wall, and escutcheon. The waste line for size 2 shall include a combination drain valve and overflow, necessary fittings to floor, and escutcheon. Each size shall have a ¾-inch I.P.S. bell-nozzle inlet assembly at top of tank, a minimum of 1 inch above the tank

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overflow, with pipe and fittings for connection to thermostatic water-mixing valve. The overflow shall be located at one end of the tank in such a manner that when the tank is filled to a point approximately 4 inches below the top rim, the overflow of water will take place.

3.5 Thermostatic water-mixing valve. — The tank shall be furnished with a water temperature control assembly consisting of one thermostatic water-mixing valve having a delivery capacity of 10 gallons (for size 1) and 20 gallons (for size 2) of water per minute at not less than 40 pounds per square inch water pressure, and shall have a temperature range from cold water minimum to not less than 120° F. maximum. The thermostatic water-mixing valve shall be equipped with a ½-inch (for size 1) or ¾-inch (for size 2) I.P.S. loose-key inlet stops, checks, escutcheons, volume control valve, vacuum breaker, pipe fittings from wall to tank, and wall mounting bracket.

3.6 Dial thermometer. — A quick-acting dial thermometer shall be mounted in the tank to permit easy and constant reading of the water temperature. Casing of dial thermometer shall be of chromium-plated brass, not less than 3½ inches in diameter. The temperature range of the dial thermometer shall be from 40° to 140° F. The capillary tube and bulb shall have a combined minimum length of 17 inches (for size 1) and 28 inches (for size 2) enclosed in a ⅜-inch-diameter brass tube. Tube shall be perforated to permit circulation of water around the bulb. Tube shall be clamped to inside of the tank at the head end.

3.7 Electric turbine ejector and aerator. — The tank shall be equipped with an electric turbine ejector and shall be operated by not less than a ⅛-horse-power, constant-speed, 115-volt, 60-cycle, 3450 r.p.m., single-phase, split-phase, vertically mounted motor requiring no condenser, and equipped with sealed-in, lifetime grease-packed ball bearings. The motor shall be of the type not requiring lubrication. The bottom-end bell shall be flat and machined to fit the motor suspension bracket in order to insure accurate alignment of the end housings of the motor as well as the turbine shaft bearing. The motor suspension bracket shall be constructed of polished aluminum or bronze. The motor shall be protected against overloads by a built-in manual-reset thermal cutout. The motor shall be equipped with a 10-ampere double-pole single-throw "on and off" switch built into a terminal box mounted on body of motor and shall be furnished with not less than a 9-foot three-conductor No. 18 heavy-duty type SJ rubber, or equivalent synthetic-rubber, covered cord and plug. If desired, the switch may be mounted in an aluminum switch box which is mounted in a fixed stationary position on one end of the tank. The switch shall be keyed in with its mounting to prevent it from rotating. A sufficient length of three-conductor No. 18 heavy-duty type SJ rubber-covered cord to permit free adjustment of the turbine position shall be furnished together with an approved strain-relief fitting to connect cord to switch box. Provision for grounding third wire of cord tc bath and bath to electrical system shall be made. The motor shall be completely covered by a chrome-plated or polished-aluminum shroud. A stainless-steel shaft of not less than ½-inch diameter, enclosed in a brass housing shall be directly connected to the motor extension shaft by a removable brass coupling and shall extend downward to revolve in a suitable molded-rubber impeller-wheel bearing mounted in the bottom housing. Bearings shall be water-lubricated, shall be provided with suitable lubricating grooves, and shall be formed of Buna-N-rubber, oil-resisting cutless bearing, or equal. The shaft shall be connected to a bronze impeller wheel which shall be enclosed in a heavy bronze housing. The turbine ejector shall deliver at maximum pressure, not less than 45 gallons of water per minute at a velocity of not less than 2500 linear feet per minute, which shall be controlled to produce an underwater stream of varying degrees of intensity from zero to maximum, and the stream shall be fan-

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shaped with a spread of not less than 8 inches at a distance of 12 inches from the ejection nozzle at maximum velocity. An aerator embodied in the turbine ejector shall deliver not less than 19,000 cubic inches per minute of heated forced air derived from thermal losses of motor, to help maintain temperature of the water. The aerating tube and elbow shall be made of brass or bronze.

3.8 Turbine-ejector elevator.—The turbine raising and lowering device shall consist of two sturdy telescoping brass tubes, the outer one attached to the tank by means of two brass or bronze brackets which shall form the guide for raising and lowering the turbine ejector. The upper bracket shall have an integral hand clamp to permit locking of the inner tube and to make adjustments for position of the turbine at higher or lower levels, as desired. To facilitate and expedite such adjustments, the inner tube shall be supported by a substantial spring which shall counterbalance the entire turbine ejector in such a fashion as to require a slight downward pressure for low-level adjustments and a slight upward pressure for high-level adjustments. The inner tube shall fit the motor suspension bracket so as to permit horizontal rotation of turbine ejector to any angle within the tank. All castings shall be made of bronze or brass tubes.

3.9 Plumbing.—All unions and connections shall be properly fitted and capable of being broken and remade without causing damage or leaks.

3.10 Finish. — All exposed sheet-metal parts, including the inside of tank, shall have a No. 4 finish. All piping, fixtures, fittings, etc., shall be chromium-plated over nickel, and exposed surfaces shall have a highly polished finish.

3.11 Safety.—The unit shall conform to all applicable requirements of the National Electrical Code.

3.12 Instructions. — The contractor shall forward to each consignee two sets of all necessary rough-in drawings for each unit upon receipt of the contract, or as soon thereafter as practicable upon receipt of delivery instructions. In addition, one set of drawings shall be securely tacked to the shipping container, and another set with a book of instructions shall be placed inside.

3.13 Service data. — The contractor shall enclose with each unit two copies of one or more booklets which contain complete instructions for installation, operation and maintenance, and a list of spare parts. Wiring diagrams shall be included for electrical items or components. Each part other than common fasteners, etc., shall be identified by the manufacturer's part number. In addition, the contractor shall mail copy of the information to Specifications Branch, Armed Services Medical Procurement Agency, 84 Sands Street, Brooklyn 1, N. Y., and one copy to Medical Technical Maintenance Division, Armed Services Medical Procurement Agency, St. Louis 2, Mo.

3.14 Workmanship. — Workmanship shall be first class throughout. Baths shall be free from defects which detract from their appearance or which may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Unless otherwise specified, at least 2 whirlpool baths of each size shall be taken at random from each 100 or fraction thereof delivered.

4.2 Inspection.—Samples shall be inspected for material, workmanship, dimensions, finish, packaging, packing, and marking.

4.3 Tests.

4.3.1 *Welded seams of tanks.*—The welded seams of the tanks shall be tested as required in **3.3.1.**

5. PREPARATION FOR DELIVERY

5.1 Packaging. — Unless otherwise specified, each unit shall be packaged in accordance with accepted commercial practice.

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5.2 **Packing.**—The subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 **Marking.**

5.3.1 *Packages.* — Unless otherwise specified, packages shall be marked with the name and size of the commodity, the quantity contained therein, and the name of the manufacturer, also the name of the contractor if different from the manufacturer.

5.3.2 *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked with the name and size of the commodity as defined by the contract or order under which the shipment is made, the quantity contained therein, the name of the contractor, the number of the contract or order, the name and address of the consignee, and the gross weight.

6. NOTES

6.1 Purchasers should specify size required, and should exercise any desired options offered herein. (See 1.2, 4.1, 5.1, 5.3, 6.2, and 7.2.5.)

6.2 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.3 This specification covers the following items appearing in the Armed Services Catalog of Medical Matériel:

7-103-005—Bath, Arm, Whirlpool, CRM, 110 Volt, 60 Cycle, AC.
7-104-005—Bath, Leg, Whirlpool, CRM, 110 Volt, 60 Cycle, AC.

6.4 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the type, etc., normally procured for Federal Government requirements.

6.5 **Transportation description.**

6.5.1 *Transportation description applicable to this item is:*

Electro-therapeutic apparatus, not otherwise indexed by name.
Carload weight 20,000 pounds subject to Rule 34, Uniform Freight Classification.
Truckload volume weight, 20,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications, and standard, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 **Army, Navy, and Air Force.**

7.2.1 *Applicable specifications and standard.*

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Military Specifications:

JAN-P-105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.

JAN-P-106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.

MIL-B-107—Boxes, Wood, Wirebound (Overseas Type).

JAN-P-125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.

Navy Department Specification:

General Specifications for Inspection of Materials.[1]

Military Standard:

MIL-STD-129—Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Packaging.* — One bath with accessories, as specified, constitutes one unit. Each unit shall be disassembled as far as is practicable. All exterior finished surfaces shall be protected by wrapping with kraft paper and cushioning with excelsior pads of flexible corrugated fiberboard or other suitable material. Wrapping and cushioning material shall be secured by tying with cotton cord or tape, or by sealing with kraft gummed paper tape. Work shall be accomplished in such a manner as to insure against marring, scratching, or damage to the unit. All accessories and disassembled parts shall be protected by wrapping with kraft paper or other suitable material and further packaged in one or more containers of suitable size and design.

7.2.3 *Packing for domestic or overseas shipment.*—Each unit shall be packed in a shipping container designed for type 3 load and conforming to the requirements of Military Specification JAN-P-105, JAN-P-106, or MIL-B-107. Closure and strapping shall conform to the requirements of appendix to applicable box specification. Each box shall be lined with a waterproof barrier fabricated of material conforming to the requirements of Military Specification JAN-P-125 and sealed with a suitable waterproof adhesive. In addition to the regular specification requirements, the shipping container design shall include skids of appropriate size to facilitate handling and to maintain in an upright position. The unit shall be adequately secured within the shipping container by means of bolts, wood braces, blocks, or other suitable devices, to insure against shifting and breakage, or marring of surfaces.

7.2.4 *Marking.*

7.2.4.1 *Interior packages.* — Each interior package shall be identified as to contents.

7.2.4.2 *Shipping containers.* — Each shipping container shall be marked in accordance with Military Standard MIL-STD-129.

7.2.5 *Inspection procedures (Navy only).* —The general inspection procedures shall be in accordance with Navy Department General Specifications for Inspection of Material.

ARMY INTEREST: M

NAVY INTEREST: Md

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog.

Copies of this specification can be purchased for 5 cents each from the General Services Administration, Business Service Center, Region 3, Seventh and D Streets, S. W., Washington 25, D. C. Single copies required for bidding purposes are available without charge at the nearest GSA Regional Office.

[1] Applicable to Navy purchases only.

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